

Teleporting Split Body: Can Portals Extend Self-Location?

Ryota Kondo ^{1*}Maki Sugimoto ²

Keio University, Japan



Figure 1: Virtual body conditions

ABSTRACT

Self-location is extended by inducing body ownership in a virtual body split into left and right. However, since it is known that body ownership of the split body is weaker than a normal body and body ownership is not induced in a body outside the peripersonal space, it is possible that if the split body moves to a distant location, the body ownership is completely lost, and self-location cannot be extended. In this study, we investigated whether using portals to maintain a body connection would generate body ownership and extend self-location in the split and distant body. The results showed that body ownership was induced in the split and distant body by presenting the portals, but the self-location was not extended.

Keywords: Body ownership, self-location, human augmentation, portals

Index Terms: I.3.7 [Computer graphics]: Three-Dimensional Graphics and Realism—Virtual Reality; H.1.2 [Models and Principles]: User/Machine Systems—human factors

1 INTRODUCTION

When we are asked, "Where are you now?" we will answer that we are here, where our innate body is located. However, this sense of being here (sense of self-location) can easily change. In virtual reality, a virtual body in front of an observer and the observer's back are simultaneously stroked, making the virtual body in front of the observer feel as if it were his or her own body (body ownership [1]) [2][3]. Then, when body ownership is induced in the virtual body in front of the observer, the sense of self-location drifts in the direction of that body [2][3]. In addition, splitting the virtual body from the center to the left or right and inducing body ownership in the virtual body and self-location is extended [4]. However, since it is reported that splitting a body [4] or a body outside the

peripersonal space (PPS) weakens body ownership [5], there is a possibility that if the virtual body is split farther away, the body ownership is completely lost and self-location cannot be extended.

On the other hand, body ownership also occurs outside the PPS when a body is connected, as in the case of the very long arm illusion [6]. Body ownership was also found to elicit when discontinuities were occluded [7] or when invisible areas were subjectively complemented [8][9], suggesting the importance of complementing body connection in these studies. Based on these studies, we hypothesized that body ownership would occur even when the body is outside the PPS if the body connections can be perceived. This would allow the extension of the self-body to places where body ownership and a sense of self-location do generally not occur.

To complement the body connection, we used portals. By using portals, body ownership in the distant virtual hand is maintained [10]. We believe that body connection can be perceived by observing a virtual body entering a portal and a virtual body exiting another portal. This study aimed to investigate whether using portals to maintain the connection of the body would generate body ownership and extend self-location to the split distant body. We expected that the portals would maintain body ownership at distances where it is normally attenuated.

2 RELATED WORKS

2.1 Body Ownership and Self-Location

We have a sense that our innate body, which we see when we look down, is our own body, which is called body ownership [1]. Then, body ownership can also be induced in other bodies than our innate body. In the rubber hand illusion [11][12], an observer's hand and a rubber hand are simultaneously stroked with a brush, and by observing only the rubber hand, the observer feels as if the rubber hand were his or her own hand. When the observer feels the rubber hand as if it were his or her own hand, pointing at the position of his or her hand from under the desk with eyes closed, the pointing position drifts in the direction of the rubber hand.

The same phenomenon can be observed with virtual bodies. By simultaneously stimulating the back of the virtual body presented in front of an observer and the observer's back, the observer feels as if the virtual body in front of him/her were his/her own body [2][3]. In Nakul et al. [3], after inducing body ownership, a ball rolls in the observer's direction from the front while the body is hidden from view. While the ball is rolling, the screen goes dark,

^{1*} ryota.kondo@keio.jp

² maki.sugimoto@keio.jp

and the observer is asked to press a button when he or she feels the ball hit his or her body. When body ownership is induced, the observer stops the ball before it reaches his or her body's original position. That is, the observer feels as if his or her body is in a more forward position than the actual position.

Maseli & Slater [5] measured body ownership and drift in self-location by applying visual-tactile stimuli to a virtual body shifted 0, 25, or 80 cm to the left from the observer. The results showed that body ownership was weaker in the 80-cm condition than in the 0- or 25-cm conditions. The drift in self-location was greater in the 25cm condition than in the 80cm condition. This suggests that lateral self-position drift does not occur when the virtual body is outside the PPS. On the other hand, in the forward directional body shift, body ownership and drift in self-location occurred even 2 m in front of the body [2][3]. In the rubber band illusion, the limit of distance that produces body ownership differs depending on the direction [13], and the same phenomenon is likely to occur in body ownership for a full body. Thus, since there is a distance limit in extending our self-location (especially in the lateral direction), we hypothesized that this limit could be exceeded by using portals.

2.2 Portals in virtual reality

Portals are gates that connect space to space. Although this study is an attempt to send a part of the body to a distant location with portals, many studies have aimed to bring space closer to the user's side. For example, placing a portal near a distant object connects it to a portal near an observer, allowing access to the distant object [14]. In a similar study, when an object is selected with a laser pointer, a portal connects to the vicinity of that object [15]. Thus, portals have been used to connect space to space and allow users to easily access distant spaces. On the other hand, Ban et al. [10] showed that using portals to teleport the virtual hand farther away resulted in stronger body ownership than the condition in which the hand was teleported without portals. This suggests that complementary body connections are important for maintaining body ownership.

2.3 Extending self-location using multiple bodies

Attempts to extend self-location have also been conducted with multiple bodies. Guterstam & Ehrsson [16] placed two mannequins in different locations and temporally switched between an image from the viewpoint of mannequin A, an image from the viewpoint of mannequin B, and an image from a bird's-eye view. Both mannequins were touched with a stick, and an observer's body was also touched with a stick at the same time. As a result, the observer felt that both mannequins were their own bodies and felt as if they were in the position of both mannequins. In a study using a robot [17], an observer manipulated the robot and saw themselves manipulating the robot from the robot's perspective at the same time, which generated the sense of owning both the robot and the innate body, and they felt as if they were in both body positions. Thus, by inducing ownership of multiple bodies, self-location is extended, but the sensation is weak and it is a subjective report only. In the present study, we aim to maintain body ownership and induce a strong sense of self-location in a distant position using portals. Also, we use behavioral measurements to investigate the sense of self-location.

3 METHODS

3.1 Participants

Twenty-one volunteers participated in the experiment. The sample size was calculated using G*Power3.1 (repeated measures ANOVA, five virtual body conditions, medium effect size $f = 0.25$, $\alpha = 0.05$, power $(1 - \beta) = 0.8$). They had a healthy vision and motor abilities. The study was conducted with the approval of the Ethical Committee on Human-subject Studies of Keio University.

3.2 Apparatus

The participants wore a motion capture suit, a head-mounted display (HMD: HTC Vive Pro Eye, 2880 x 1600 pixel, 90Hz, 110deg), and two Vive controllers (Figure 2). They observed a computer (OS: Windows 10, CPU: AMD Ryzen 9 5900X 12-Core Processor 3.70 GHz, RAM: 16GB, GPU: NVIDIA GeForce RTX 2080) generated virtual environment using Unity. The participants' movements were tracked by 24 Optitrack cameras (fourteen PrimeX 22 cameras: 2048 x 1088 pixel, 120Hz; six Prime 17W cameras: 1664 x 1088 pixel, 120Hz; four Prime 13 cameras: 1280 x 1024 pixel, 120Hz).



Figure 2: Experimental setup

3.3 Stimuli and conditions

There were five virtual body conditions: Normal, Portal25, Portal80, Split25, and Split80 (Figure 1). In Normal condition, a normal virtual body was presented. In the portal conditions, a normal virtual body and portals were presented. The distance between the portals was 25cm (Portal25) or 80cm (Portal80). When the participant inserted his or her virtual body into a portal, the body part he or she inserted appeared from another portal. The viewpoint did not move to another portal location when the head entered a portal. In the split conditions, a virtual body was split left and right, and the left side of the body shifted 25cm (Split25) or 80cm (Split80) to the left. Portal distances and splitting distances were determined based on Maselli and Slater [5]. This is because the 25cm shift in their study generated the same level of body ownership as the 0cm shift, and the 80cm shift resulted in weaker body ownership. These virtual bodies were moved synchronously with the participant's movement. The participants observed the virtual body from the first-person perspective through the HMD.

3.4 Learning session

The participants were asked to touch a target on the left side of the participant's virtual body with their hands and feet to induce body ownership (Figure 3). There were two targets: a shoulder height bar touched with the hand and a line on the ground that was touched with the foot. After touching the two targets, the participant stepped on the line of the initial position with the left foot of the virtual body and then touched the targets on the left side of the virtual body again. The target that the participant should touch was presented in red, and when the participant touched the target, it turned green. This task was repeated for 2 minutes.

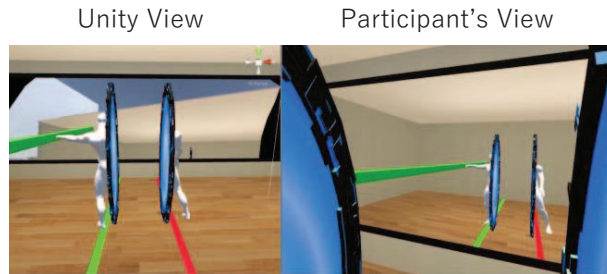


Figure 3: Picture of learning session

3.5 Mental imagery task

We used the mental imagery task (MIT) to measure self-location based on Nakul et al. [3]. In the task, the ball's direction was indicated by letters on a darkened screen, and the participants were instructed to turn their heads only in the direction indicated. Then, when the participant pressed a button, an image of the room was shown, and when the button was pressed again, a virtual ball rolled toward the participant. After the ball started to move for 3 s, the screen turned dark. The participants were instructed to imagine the ball's movement even after the darkening and to press the button on the controller when they felt the ball hit their right foot if it rolled from the right or their left foot if it rolled from the left. This position was recorded as the estimated self-location of the foot. The MIT was conducted in random order six times on each side (Figure 4).

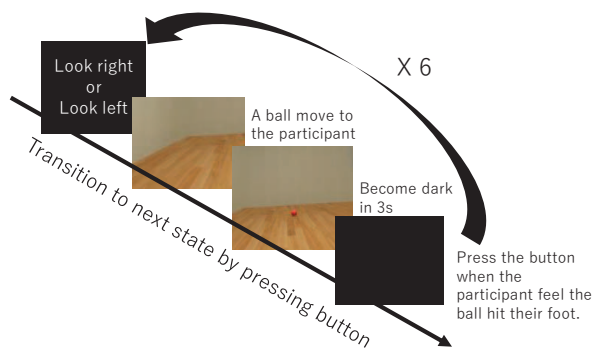


Figure 4: Procedure of mental imagery task

3.6 Procedures

In the experiment, the participants first practiced the MIT and then performed the MIT again as a baseline. The participants performed the learning task for 2 minutes and then performed MIT. Finally, they answered a questionnaire about body perception on a seven-level Likert scale (-3: strongly disagree, +3: strongly agree). The procedure from the learning session to the questionnaire was performed in random order five times, once for each condition (five virtual body conditions x one repetition, Figure 5).

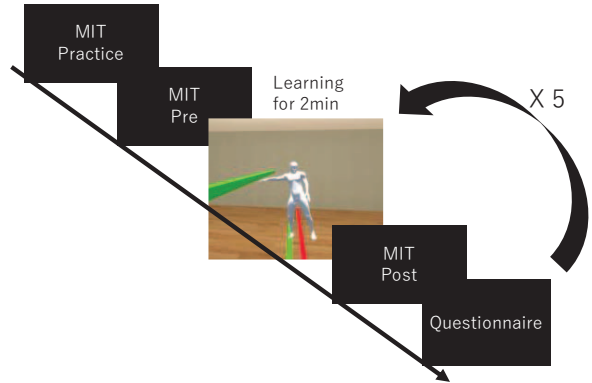


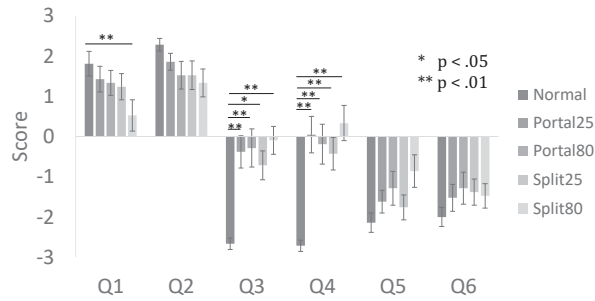
Figure 5: Procedure of experiment

4 RESULTS

We used JASP for the statistical analysis. In the MIT analysis, one participant with trials that did not press a button was excluded. Freedman's test was performed for the questionnaire results (Figure 6) and MIT results (Figure 7). Conover's test was used for the post-hoc test. The p -values were corrected by the Bonferroni method.

4.1 Questionnaire

There was a main effect in Q1 ($p = .005$). Body ownership of the Normal condition was higher than the Split80 condition (Normal vs. Split80: $p = .005$). In the questionnaire of agency (Q2), there was no main effect ($p = .11$). In Q3, a main effect was found ($p < .001$), and the score was higher in the Portal25, Portal80, Split25, Split80 condition than in the Normal condition (Normal vs. Portal25: $p = .001$; Normal vs. Portal80: $p < .001$; Normal vs. Split25: $p = .012$; Normal vs. Split80: $p < .001$). However, the score was not significantly higher than 0 in the Portal25 ($p = .34$), Portal80 ($p = .50$), and Split80 ($p = .66$) conditions. Also, the score was significantly lower than 0 in the Split25 condition. A main effect was found in Q4 ($p < .001$), and the participants felt as if their body was split in the Portal25, Portal80, Split25, and Split80 condition than Normal condition (Normal vs. Portal25: $p < .001$; Normal vs. Portal80: $p = .002$; Normal vs. Split25: $p = .007$; Normal vs. Split80: $p < .001$). On the other hand, the score was not significantly higher than 0 in the Portal25 ($p = .79$), Portal80 ($p = .65$), Split25 ($p = .26$), and Split80 ($p = .45$) conditions. The control items had no main effect (Q5: $p = .068$; Q6: $p = .33$).



Q1	I felt as if the virtual body was my body.
Q2	The movements of the virtual body seemed to be my movements.
Q3	I felt as if I had two bodies.
Q4	I felt as if my body had split.
Q5	I felt like I had no body.
Q6	I felt as if the virtual body was controlling my movements.

Figure 6: Results of questionnaire. Error bars indicate SE.

4.2 Mental Imagery task

The values for each condition minus the baseline value to eliminate individual differences. The results showed no main effect of the virtual body condition on the left side of MIT ($p = .39$) and the right side of MIT ($p = .71$), but rightward drift in the Split80 condition was significantly greater than 0 ($p = .044$). This means that the participant's perceived right foot position drifted significantly to the right.

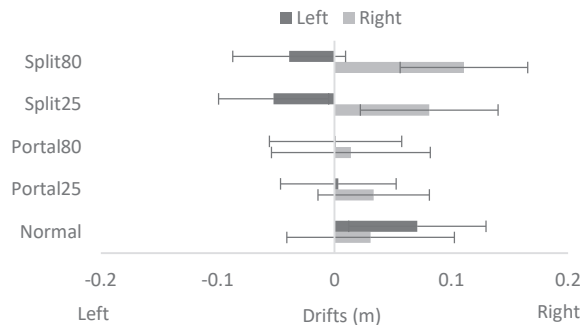


Figure 7: Results of mental imagery task. Error bars indicate SE.

5 DISCUSSION

5.1 Body Ownership and Self-Location

In this study, we investigated whether using portals to maintain the connection of the body would generate body ownership and extend self-location to the split distant body. As a result, body ownership of the far split body (Split80) was weaker than in the Normal condition. On the other hand, there was no significant difference in body ownership in the far portal condition (Portal80) compared to the Normal condition. Thus, body ownership was maintained for the distant body by presenting the portals.

Based on the results of the previous study [5], drift in self-location was expected under the Split25 and Portal25 conditions,

which are shifts within the PPS. We also expected that the portals would cause self-location drift even at a distance of 80 cm, where drift does not normally occur. However, there was no significant difference between any of the conditions at MIT. There is a wide variation in the results, possibly because each participant has a different skill level in the task. Our results suggest that portals can maintain body ownership over a distant body but cannot extend self-location.

5.2 Drift in self-location to the opposite direction

Although there were no differences between conditions, the self-location of the right foot shifted to the right in the Split80 condition. While touching the target on the left side, some participants moved too far to the left in the Split and Portal conditions. This movement was likely the result of the participants' body schema not being updated to match the shifted body to the left and attempting to reach the target from the participant's body position. Then, to compensate for the shift, the body schema was probably updated in the direction away from the target on the left side rather than in the direction of the left side of the body. Thus, in the Split80 condition, the participants felt that the left side of the split body was another body following their innate body, and their self-location may have shifted to the right side to touch the target with that half of the body accurately. It is supported by the fact that in the subjective ratings, only the Split80 condition showed significantly weaker body ownership than the Normal condition.

5.3 Sense of having multiple bodies and sense of split

The score for Q3 (I felt as if I had two bodies) was significantly higher in the portal and split conditions than in the Normal condition, but the score was not significantly higher than 0 (Portal25, Portal80, Split25, Split80). The score was significantly lower than 0 in the Split25 condition. Thus, even in the case of a split, the participants did not feel as if they had two bodies. Although it has been reported that a split as close as 25 cm is perceived as one body [4], a similar tendency was observed in the condition with an 80 cm shift. It is possible that both split bodies moved synchronously when the body was tilted and that this caused the perception of integration as a single body. In the portal condition, the participants did not feel as if they had two bodies as a result of the perceived connection of the body due to the portal. In Q4 (I felt as if my body had split), as in Q3, the participants did not feel as if they were split so much, probably because they perceived the virtual body as one body in the portal and split conditions.

6 LIMITATIONS

Since the previous portal studies [14][15] had moved the space closer to an observer, it is possible that drift in self-location did not occur because the participants felt as if the space was coming closer to them rather than that their bodies had moved farther away by the portals. Further investigation is needed to determine whether the participants felt their bodies moved farther away or the space moved closer to them.

In the experiment, the participants were instructed to repeat the action of entering and exiting the portals to make them aware that the portals connected their bodies. The movement back out of the portal may have temporarily created the same condition as the Normal condition, which may have prevented the drift in self-location. Additionally, this may be why body ownership did not decrease in the portal conditions.

The viewpoint remained fixed even after passing through the portals. Because the viewpoint did not move in the study of Maseli & Slater [5] and the split body study [4]. However, there is a high possibility that the position of the viewpoint is felt as the position

where one is located, and it may be possible to extend the self-location by manipulating the position of the viewpoint.

7 CONCLUSION

In this study, we investigated whether using portals to complement body connections could generate a sense of body ownership and extend self-location farther away from the divided, distant body. The results showed that the portals generated body ownership even in the distant, split body. On the other hand, the portals did not extend self-location. These results suggest that portals can maintain body ownership over distant bodies but cannot extend self-location.

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